Formula One Race Outcome Prediction

Exploratory Data Analysis & Feature Engineering

MSDS 422: Practical Machine Learning

Authors: Sara Alsiyat · Qifan Yang · Boqi Niu

Professor: Dr. Irene Tsapara

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Problem Statement & Research Objectives

The Problem

F1 race outcomes depend on driver skill, constructor performance, track characteristics, and race conditions.

Finishing in the top 10 determines whether a driver earns championship points, a critical threshold for competitive success.

Can machine learning predict top-10 finishes using only pre-race information?

Research Objectives

- Build and evaluate ML models for top-10 finish prediction using historical F1 data
- Examine how qualifying position, constructor, driver history, and circuit affect outcomes
- Explore circuit-level variation in race unpredictability
- Compare models for accuracy, interpretability, and decision-support value

Literature Review

Key findings from 9 reviewed studies informing our approach

Rockerbie & Easton (2022)	Driver–constructor interaction explains large portion of race performance; both factors must be modeled jointly.
van Kesteren & Bergkamp (2023)	Bayesian analysis shows ~88% of F1 result variance from constructor; driver skill accounts for minority.
Bansal (2025)	XGBoost achieves $R^2 = 0.999$ on championship points; grid position contributes 75.8% of model performance.
Patil et al. (2023)	Pit frequency and tyre strategy significantly affect championship points; PCA captures 70% variance in 4 components.
Urdhwareshe (2025)	TabNet achieves $R^2 = 0.75$ on driver positions using only pre-race features; interpretability via attention masks.
Jafri (2024)	LSTM-based models capture temporal momentum in lap times; composite loss functions improve positional accuracy.
Gu et al. (2021)	NeuralAC models cooperation/competition effects; interaction features outperform individual-only approaches.
Zheng (2022)	Combining internal performance + external context (Elo, momentum) improves sports outcome prediction.

Full annotated bibliography with 9 papers available in the written report.

Dataset & Integration Strategy

1950–2024	26,000+	11	80+
Time Span	Driver–Race Records	Tables Merged	Features Created

Tables Integrated

Results	Races	Drivers	Constructors
Circuits	Qualifying	Status	Pit Stops
Lap Times	Driver Standings	Constructor Standings	

Target Variable: top10_finish — binary indicator (1 = driver finished in top 10, 0 = otherwise). Derived from positionOrder ≤ 10.

Key EDA Insights

Grid Position Dominance

Starting position is the strongest predictor of top-10 finishes. Pole position converts to top-10 at >95% rate; grid 15+ drops below 30%.

Constructor Effect: 88%

Van Kesteren & Bergkamp (2023) found ~88% of F1 result variance is attributable to the constructor, not the driver.

DNF Rate Decline

Did Not Finish rates have dropped from ~40% in the 1950s to under 10% in recent seasons, reflecting reliability improvements.

Circuit Variation

Some circuits show much higher position volatility, suggesting more unpredictable races — important for modeling uncertainty.

Feature Engineering: 35+ Features Across 12 Categories

Temporal

Driver age, race month, season stage

Performance History

Rolling 5-race top-10 rate, DNF rate, avg position

Constructor

Team rolling stats, season top-10 rate

Circuit

DNF rate, volatility, driver-circuit history

Driver–Team Interaction

Races together, joint success rate

Qualifying

Grid position, front-row start, top-5/top-10 flags

Competitive Context

Field size, grid percentile

Momentum

Top-10 streak, points last 3, position trend

Pit Stop

Avg stops per race, constructor pit efficiency

Championship

Standings position, points gap to leader

Era

Regulation era encoding (1950s to ground effect)

Categorical

Label-encoded driver, constructor, circuit, country

Data Leakage: The Issue & The Fix

What Went Wrong

Initial models scored 1.000 ROC AUC — a clear sign of data leakage.

Post-race features were included as model inputs:

- positionOrder — directly defines the target
- points — only awarded to top-10 finishers
- milliseconds, laps — post-race outcome data
- Championship standings included current race

How We Fixed It

Strict Leakage Guard
Explicit blocklist of 12+ post-race columns removed from feature matrix.

Temporal shift(1) on All Rolling Features
Every expanding/rolling window excludes the current race.

Championship Standings Fix
Shifted to previous round's standings so model only sees pre-race info.

Leakage Audit Cell
Automated check flags any feature with |correlation| > 0.85 to the target.

Baseline Model Performance

Temporal 3-way split: Train (< 2018) | Validation (2018–2021) | Test Holdout (2022–2024)

Validation Set (2018–2021)

Model	ROC AUC	PR AUC	F1
Logistic Regression	0.822	0.794	0.727
Random Forest	0.840	0.802	0.781
Gradient Boosting	**0.827**	**0.801**	**0.755**

Test Holdout (2022–2024)

Model	ROC AUC	PR AUC	F1
Logistic Regression	0.857	0.849	0.775
Random Forest	**0.869**	**0.854**	**0.808**
Gradient Boosting	0.858	0.833	0.796

Key Takeaways

- Random Forest achieves best overall performance (0.869 ROC AUC on holdout)
- Test scores slightly higher than validation — no overfitting detected
- All models use only pre-race features after leakage fix — results are realistic

Implementation Strategy & Pipeline

1

Data Collection

Kaggle F1
11 CSV tables

2

Data Integration

Merge into
driver–race rows

3

EDA

Patterns,
missing data

4

Feature Engineering

35+ features
shift(1) safe

5

Leakage Audit

Block post-race
columns

6

Baseline Models

LR, RF, GB
3-way split

7

Model Selection

MLP
hyperparameter search

8

Ensemble Learning

Voting, stacking
weighted blend

9

SHAP Analysis

Feature importance
interpretability

10

Deployment

Registry, batch
monitoring

Key Design Principles

Reproducible pipeline: EDA → feature engineering → modeling → evaluation → deployment

Validation-first mindset: Model choices based on validation; test set held out for final confirmation

Full Model Comparison

Temporal 3-way split: Train (< 2018) | Validation (2018–2021) | Test Holdout (2022–2024)

Validation Set (2018–2021)

Model	ROC AUC	PR AUC	F1
Baseline (Grid LR)	0.800	0.745	0.748
Logistic Regression	0.821	0.793	0.758
MLP	0.830	0.803	0.770
Gradient Boosting	**0.832**	**0.805**	**0.764**
Random Forest	0.837	0.805	0.763

Test Holdout (2022–2024)

Model	ROC AUC	PR AUC	F1
Baseline (Grid LR)	0.790	0.724	0.757
Logistic Regression	0.856	0.848	0.786
MLP	0.856	0.835	0.793
Random Forest	0.863	0.832	0.790
Gradient Boosting	**0.864**	**0.852**	**0.789**

Key Takeaways

- Gradient Boosting achieves the best PR AUC on both validation (0.805) and test (0.852)
- All five models significantly outperform the grid-only baseline, validating the feature engineering pipeline
- Test scores are consistent with validation — no overfitting detected across the temporal split
- Tree-based models (RF, GB) consistently outperform linear and neural network approaches

Feature Importance & SHAP Analysis

Top Predictive Features

- Grid position (starting position) — strongest single predictor
- Constructor rolling top-10 rate — team strength signal
- Driver 5-race rolling top-10 rate — recent form
- Grid percentile — relative field position
- Championship standing position — season context
- Driver–constructor joint success rate — team synergy

Key SHAP Insights

- Grid position has highest mean |SHAP value| — confirms domain knowledge
- Constructor features show strong positive SHAP for top teams (Mercedes, Red Bull)
- Driver momentum features (streaks, recent form) show asymmetric effects
- Circuit-specific features have moderate but consistent directional influence
- Era encoding captures structural shifts in competitiveness across regulation periods

Methodology

- Permutation importance computed on test set with 8 repeats for stability
- SHAP TreeExplainer applied to best tree-based model on 1,000-sample subset
- Both methods confirm grid position and constructor strength as dominant predictive signals

Ensemble Learning Results

Four ensemble strategies built from the top 2 validation models (Random Forest, Gradient Boosting)

Ensemble Method	Val PR AUC	Test PR AUC	Test ROC AUC	Test F1
Soft Voting	0.808	0.845	0.867	0.797
Weighted Voting	0.808	0.845	0.867	0.797
Stacking (Meta-LR)	**0.809**	**0.842**	**0.868**	**0.798**
Hard Voting	0.723	0.763	0.816	0.794

Best Single Model

Gradient Boosting

- Test ROC AUC: 0.864
- Test PR AUC: 0.852
- Test F1: 0.789
- Fastest inference, simplest to deploy

Best Ensemble

Stacking (Meta-LR)

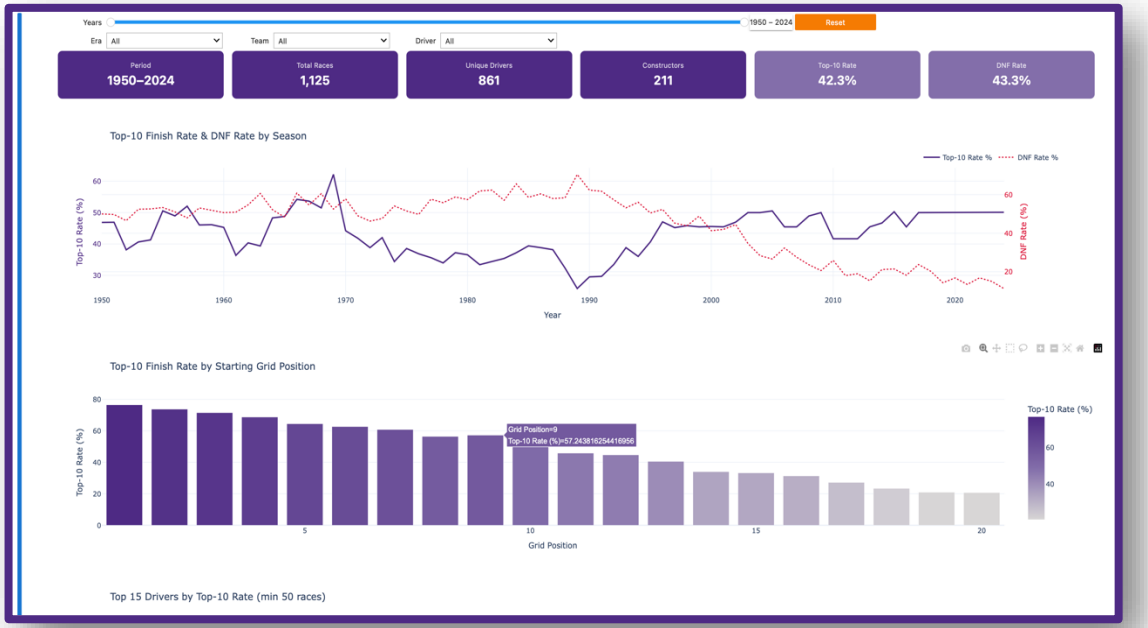
- Test ROC AUC: 0.868
- Test PR AUC: 0.842
- Test F1: 0.798
- Marginal ROC gain, lower PR AUC

Takeaway

- Ensembles match but do not clearly exceed the best single model (Gradient Boosting) on this dataset
- Gradient Boosting is recommended for deployment due to comparable performance with simpler architecture

Interactive Dashboard

Live Filtering Dashboard built with ipywidgets + Plotly inside the notebook



Note: Full interactive dashboard runs in the Jupyter notebook with filter controls at top.

Dashboard Features

Year Range Slider
Filter any period from 1950–2024

Era Dropdown
Compare regulation periods

Team Filter
Isolate constructor performance

Driver Filter
Analyze individual drivers

KPI Cards
Period, races, drivers, top-10 rate, DNF rate

7 Interactive Charts
Trends, grid position, top drivers, constructors, era comparison, circuit explorer

Model Deployment & Monitoring

1

Model Registry

Best model and scaler saved via joblib for reproducible deployment. Feature list and metadata stored as JSON artifacts.

2

BatchPredictor

Offline scoring utility that loads the serialized model and scaler, accepts a DataFrame of pre-race features, and returns top-10 probability estimates.

3

ModelMonitor

Drift detection module that tracks feature distributions and prediction statistics over time. Flags when incoming data diverges from training patterns.

4

Interactive Dashboard

Notebook-based dashboard with season, driver, and constructor filters. Visualizes predicted probabilities alongside actual outcomes for rapid model evaluation.

Business Impact & Lessons Learned

Business Impact

- Builds a predictive system estimating a driver's Top-10 finish probability before a race using historical F1 data.
- Provides data-driven insights for teams and analysts to evaluate race strategies (qualifying setup, tire choices, reliability risks).
- Enhances broadcast storytelling and race analytics by quantifying performance trends and competitive momentum.
- Supports fan engagement platforms such as fantasy sports and prediction markets through transparent probability forecasts.
- Establishes a reproducible framework for analyzing driver, constructor, and circuit performance across eras.

Lessons Learned

- Preventing data leakage is critical in sports prediction models.
- Race outcomes should not influence pre-race feature construction.
- Three factors dominate predictive power: Grid position, Constructor strength, and Recent driver performance
- Combining short-term rolling metrics with long-term career statistics improves prediction stability.
- Extensive feature engineering improves signal quality but requires strong evaluation and interpretability checks.

Assumptions & Limitations in the Analysis

Assumptions

- All engineered features represent information available before race start to prevent data leakage.
- Rolling statistics (e.g., last-5 race performance) are built using historically shifted windows.
- Top-10 finish is used as a consistent proxy for race success across different eras.
- Missing qualifying data is approximated using grid position and median imputation.
- Label encoding assumes tree-based models can learn entity differences without artificial ordering.
- Time-based data split:
- Train: <2018
- Validation: 2019–2021
- Test: 2022–2024

Limitations

- Formula One evolves over time, with rule changes, technological advances, and competitive shifts affecting model stability.
- Dataset lacks important contextual variables such as: weather conditions, safety car events, pit strategy, penalties, and tire compounds
- Early-era missing qualifying data required imputation, potentially weakening signal quality.
- Several engineered features exhibit multicollinearity, complicating interpretation.
- Repeated driver-season patterns may create residual dependence, affecting generalization.
- No weather or safety car data — key real-time variables unavailable in the dataset
- Label-encoded categorical may not capture constructor/driver relationships optimally
- Model assumes static pre-race features; in-race dynamics are excluded by design

Conclusions & Future Work

Key Findings

- Gradient Boosting achieves the best single-model performance (PR AUC 0.852, ROC AUC 0.864 on test)
- Ensemble methods match but do not clearly exceed the best single model
- Grid position and constructor strength are the dominant predictive features confirmed by SHAP
- The temporal split confirms generalization — no overfitting across a 6+ year test period
- All five models significantly outperform the grid-only baseline (PR AUC 0.724)

Future Directions

- Integrate weather and safety car data as identified by Jafri (2024)
- Explore LightGBM and TabNet for additional gradient boosting baselines
- Build a live prediction interface for pre-race probability estimates
- Extend to multi-class position prediction (e.g., podium, points zone)
- Enforce a strict pre-race feature set to avoid outcome leakage.
- Apply hyperparameter tuning and probability calibration.
- Conduct cross-era evaluation to detect performance degradation from regulatory changes.
- Integrate additional contextual data such as weather, safety cars, penalties, and tire strategies.
- Deploy the best model as a production inference pipeline (batch or API) with monitoring and periodic retraining.

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Team Contributions

Team Member	EDA	Feature Engineering	Modeling	Project Oversight	Presentation Creation	Report Writing	Dashboard	Limitations & Future Work
Sara Alsiyat	✓	✓	✓	✓	✓	✓	✓	
Qifan Yang	✓	✓	✓		✓	✓		✓
Boqi Niu	✓	✓	✓	✓	✓	✓	✓	

Thank you…

Sara Alsiyat · Qifan Yang · Boqi Niu

MSDS 422 · Dr. Irene Tsapara · Northwestern University